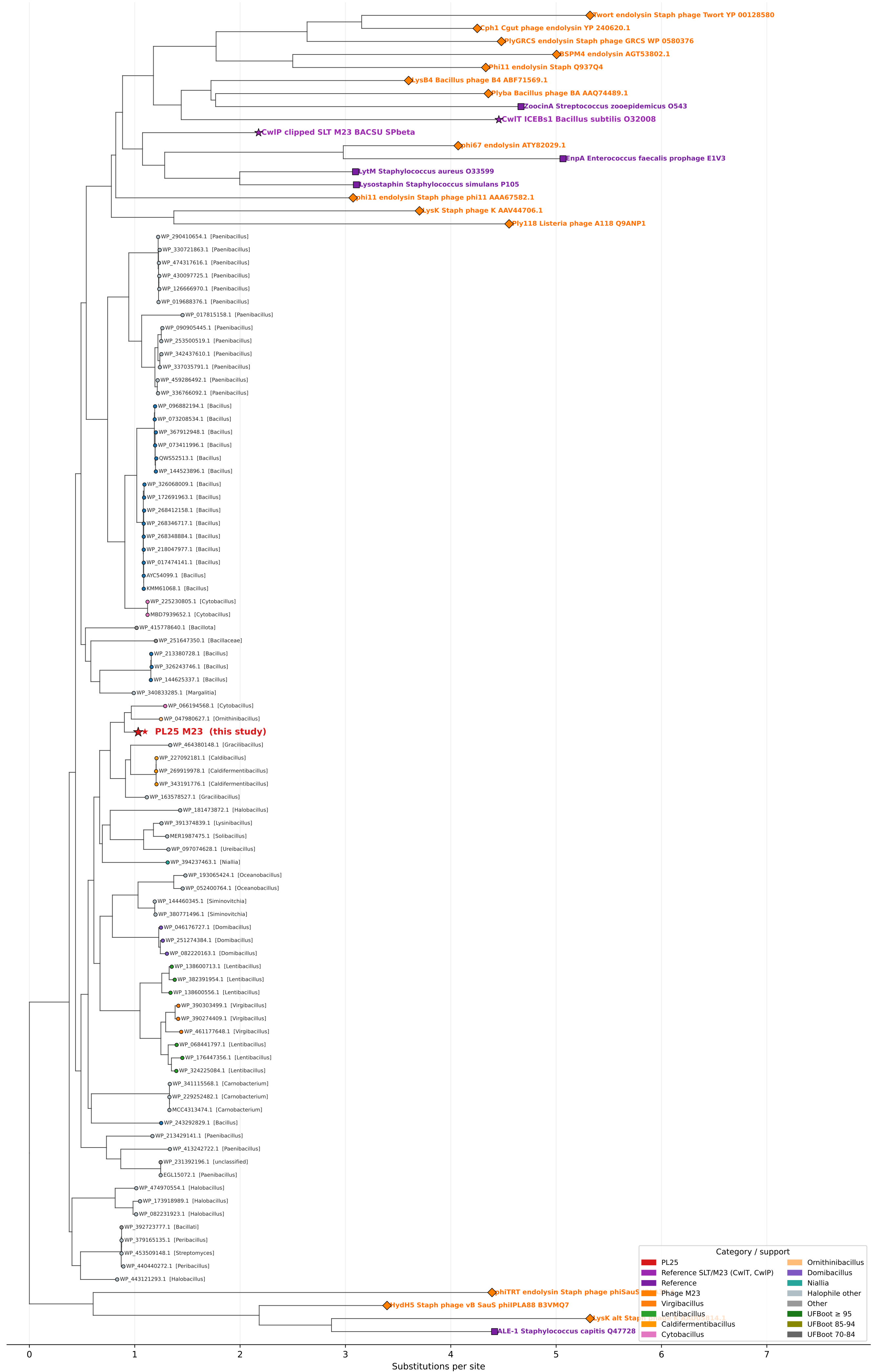


**M23 phylogeny v3 — PL25 plasmid SLT/M23 in context of CwIT (ICEBs1) and CwIP (SP-β prophage)**  
**IQ-TREE3 · Q.PFAM+G4 · 1,000 UFBoot · 102 sequences · midpoint-rooted**



red = PL25 M23 (plasmid-encoded, this study) • ★ purple = bifunctional SLT/M23 references (CwIT ICEBs1, CwIP SP- $\beta$  clipped) • □ purple-dark = other M23 reference enzymes (lysostaphin, LytM, ALE-1, EnpA, ZoocinA) • ◆ orange = phage-encoded M23 lysins • = halophile-Bacillaceae homologues from NCBI nr. PL25 M23 is highly divergent (~29% identity) from both CwIT and CwIP, defining a separate clade.